Lecture Notes on

The Roman Space Telescope

Prepared by Abir Mehta

amehta251@student.fuhsd.org dlagu234@student.fuhsd.org

These notes open the 2026–27 year. We first review the units astronomers use to talk about distance, then turn to NASA's newest instrument, the Nancy Grace Roman Space Telescope, which launched on August 30, 2026. We look at who Roman was, what the telescope is built from, how it finds planets, and why it may finally pin down dark energy.

Contents

1 How We Measure Space **3**
1.1 Light-Year . . . 3
1.2 Astronomical Unit (AU) . . . 3
1.3 Parsec . . . 3
1.4 For Scale . . . 3

2 The Nancy Grace Roman Space Telescope **3**
2.1 The Woman Behind the Telescope . . . 3
2.2 The Specs . . . 3

3 Hubble versus Roman **4**

4 What Roman Will Find **4**

5 How Roman Finds Planets: Gravitational Microlensing **4**

1 How We Measure Space

1.1 Light-Year

- A light-year is the distance light travels in one year, about 5.88×10^{12} miles.
- It is a unit of *distance*, not time.

1.2 Astronomical Unit (AU)

- 1 AU is the average distance between Earth and the Sun, about 9.3×10^{7} miles.
- Useful for describing distances inside a planetary system.

1.3 Parsec

- 1 parsec ≈ 3.26 light-years.
- This is the unit professional astronomers actually use.

1.4 For Scale

- Alpha Centauri, the nearest star system, is about 4.35 light-years away.
- The Milky Way is roughly 100,000 light-years across.

2 The Nancy Grace Roman Space Telescope

2.1 The Woman Behind the Telescope

- Nancy Grace Roman (1925–2018) became NASA's first Chief of Astronomy in 1959 and its first woman executive.
- She fought for years to get the Hubble Space Telescope funded and won, which is why she is called the **Mother of Hubble**. Roman is her telescope.

2.2 The Specs

- **Launch.** Roman launched on August 30, 2026 at 4:26 AM PDT on a SpaceX Falcon Heavy from Launch Complex 39A at Kennedy Space Center.
- **Mirror.** A 2.4 m primary mirror, the same size as Hubble's. NASA received two such mirrors for free from a retired spy-satellite program.
- **Camera.** A 300-megapixel infrared camera built from 18 detectors in a curved arc. The shape became the mission logo.
- **Location.** Parked at the second Lagrange point (L2), about 1.5×10^{6} km from Earth, right next to the James Webb Space Telescope.
- **Data.** About 1.4 TB per day, more in a single day than Hubble makes in a whole year, over a planned 5-year mission.

3 Hubble versus Roman

- A single Roman image covers a patch of sky larger than the full Moon, at the same sharpness as Hubble. A Hubble or JWST frame covers a patch about the size of a grain of salt held at arm's length.
- **Andromeda, side by side.** To image the Andromeda Galaxy, Hubble needed 400 individual images, 1,300 imaging hours, and 4 years. Roman needs 2 images and about 1 hour, and both results contain the same 1.5 billion pixels.

4 What Roman Will Find

- **Exoplanets.** About 100,000 of them, roughly 16 times more than every exoplanet humanity has found so far, from this one mission.
- **Rogue worlds.** Starless planets drifting alone through the galaxy, some as small as Mars.
- **Dark energy.** Dark energy makes up about 68% of the universe and drives its accelerating expansion, yet after 28 years no one knows what it is. Roman will map about 620 million galaxies and 10,000 exploding stars to trace how space is stretching. Recent DESI results hint dark energy may be changing over time, and Roman is expected to settle the question within 5 years.

5 How Roman Finds Planets: Gravitational Microlensing

- When a foreground star drifts in front of a more distant background star, the foreground star's gravity bends and focuses the background star's light, briefly brightening it. This is **gravitational microlensing**.
- If the foreground star has a planet, the planet adds a small extra spike to the light curve. That spike reveals worlds up to about 26,000 light-years away.
- Einstein predicted this effect in 1936 and thought it would never be observable in practice. Roman turns it into a survey tool.